INDUS VALLEY CIVILIZATION

By

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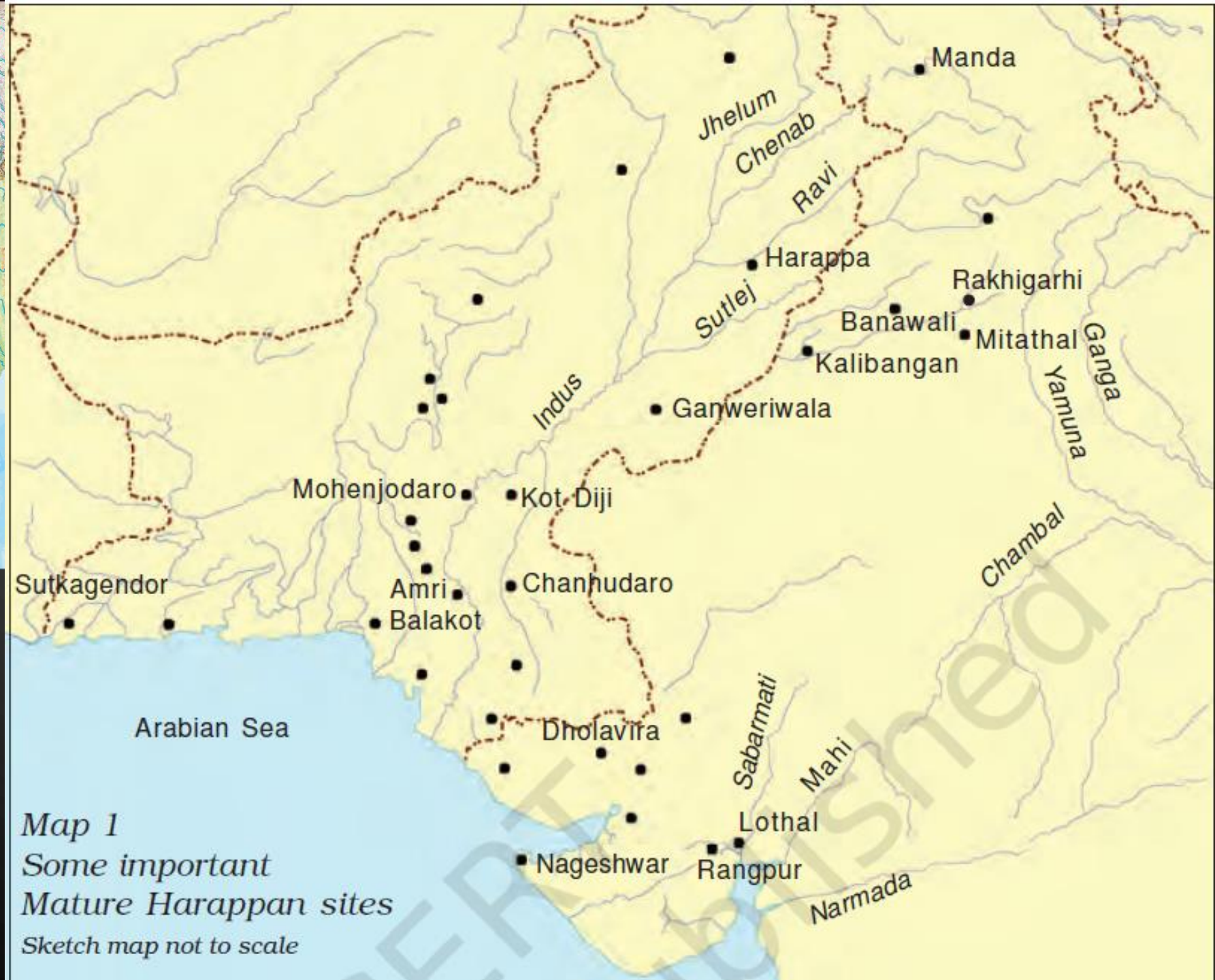

Map 1
Some important Mature Harappan sites
Sketch map not to scale

Discovery of the Indus Valley Civilization

The story of Harappa
Railway Line (1850's)

Harappa or IVC or Something else?

Sindhu–Saraswati'

Origin of Indus Valley Civilization

Indigenous Theory	John Marshall	A long antecedent history on the soil of India.
Diffusionist Theory	E. J. H. Mackay	a migration of people from Sumer (southern Mesopotamia)
Migration of Ideas	Mortimer Wheeler	Migration of Ideas instead of people

General Features of the Indus Valley Civilization

- Town Planning
- Economic Life
- Social Life
- Religious Life
- Political Life
- Language & Scripts

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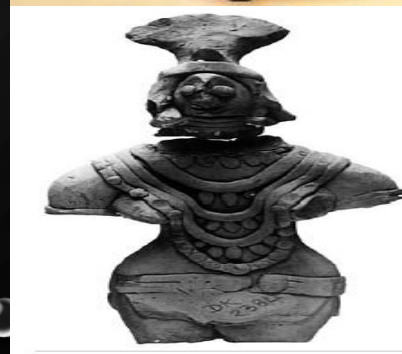


FEMALE FIGURINE WITH FAN-SHAPED HEADDRESS

Important Sites of Indus Valley Civilization

Year	Site	Location	Excavated by	Major findings
1921	Harappa	Sahiwal District, Punjab on the banks of Ravi	Daya Ram Sahni	Piece of Pottery with Indus Script
				Cubical Limestone Weight
				Faience Slag
				Sandstone statues of Human Anatomy
				Copper Bullock cart
				Granaries
				Coffin burials (Only founded in Harappa)
				Terracotta Figurines
1922	Mohenjo-Daro	Larkana, District of Sind on the bank of Indus	R. D Banerjee	Great bath
				Granary
				Unicorn Seals (Most numbers of it in here)
				Bronze dancing girl statue
				Seal of a man with deer, elephants, tigers and rhinos around- Considered to be a Pashupati Seal)
				Steatite statue of Beard man
				Bronze buffalo
1929	Sutkagendor	Baluchistan on Dast river	Stein	Trade point between Harappa and Babylon
				Flint Blades
				Stone Vessels
				Stone Arrowheads
				Shell Beads
				Pottery

Important Sites of Indus Valley Civilization

1931	Chanhudaro	Mullan Sandha, Sind on the Indus River	N G Majumdar	Bangle Factory
				InkPot
				Bead makers shop
				The footprint of a dog chasing a cat
				Cart with a seated driver
				Note: It is the only city without a citadel
1935	Amri	Close to Balochistan, on the bank of the Indus River	N G Majumdar	Antelope evidence
				Rhinoceros' evidence
1953	Kalibangan	Hanumangarh District, Rajasthan on the bank of Ghaggar river	Amlanand Ghose	Lower fortified town
				Wooden drainage
				Copper Ox
				Evidence of earthquake
				Wooden plough
				Camel's bone
				Fire alters
				Camel's bones
				Furrowed land
1953	Lothal	Gujarat on the Bhogva River near the Gulf of Cambay	R. Rao	Port Town
				Graveyard
				Ivory weight balance
				Copper dog
				First manmade port
				Dockyard
				Rice husk
				Fire alters
				Chess-playing

Important Sites of Indus Valley Civilization

1964	Surkotada	Gujarat	J P Joshi	Bones of horses
				Beads
				Stone Covered Beads
1974	Banawali	Fatehabad district of Haryana	R S Bisht	Beads
				Barley
				Oval shaped settlement
				The only city with radial streets
				Toy plough
				The largest number of barley grains
1985	Dholavira	Gujarat in Rann of Kutchchh	R S Bisht	Exclusive water management
				Only site to be divided into three parts
				Giant water reservoir
				Unique water harnessing system
				Dams
				Embankments
				Stadium
				Rock – Cut architecture

India Sculpture and Pottery (IVC)

Seals

Archaeologists have found numerous seals of different shapes and sizes all across the excavation sites. While *most of the seals are square*, it has been found that triangular, rectangular and circular seals were also being used. **Steatite**, a soft stone found in the river beds, was the most common material used to make seals, but agate, chert, copper, faience and terracotta seals were also found. Some instances of gold and ivory seals have also been found.

Most of the seals have inscriptions in a **pictographic script** which is yet to be deciphered. The script was written mostly from right to left, but **bi-directional writing style,** that is right to left on one line and left to right on another line, has also been found. *Animal impressions* were also present (generally five) which were carved intaglio on the surfaces. The common animal motifs include a unicorn, humped bull, rhinoceros, tiger, elephant, buffalo, bison, goat, markour, ibex and crocodile. However, **no evidence of cows** has been found on any seal. Generally, the seals had an animal or human figure on one side and an inscription on the opposite side or inscriptions on both the sides. Some seals had inscriptions on a **third side as well**.

Seals were **primarily used for commercial purposes** and helped in communication. Discovery of various seals in Mesopotamia and various sites such as Lothal indicate that seals were extensively used for trade. Some seals with a hole on them have been found on dead bodies indicating its **use as amulets**, carried on the persons of their owners, probably seen as some form of identification. *Mathematical images* have also been found on some seals, which might have been **used for educational purposes** as well. Seals with symbol similar to 'Swastika' design have also been found.

Famous seals include: Pashupati Seal and Unicorn Seal.

Unicorn Seal

Pashupati Seal: *It is a steatite seal discovered at Mohenjo-daro and depicts a* ***human figure*** *or a deity seated cross-legged. The figure, referred to as Pashupati, wears a three-horned headgear and is surrounded by animals. An* ***elephant*** *and a* ***tiger*** *are present on the left side of the figure while a rhinoceros and a* ***buffalo*** *are seen on the right side. Two* ***antelopes*** *are shown below the seat of the figure.*

India Sculpture and Pottery (IVC)

Bronze Figures

The Harappan Civilisation saw wide-scale practise of bronze casting. The bronze statues were made using **'lost wax technique' or 'Cire Perdue'**. In this technique, wax figures are first coated with wet clay and allowed to dry. The clay-coated figures are then heated, allowing the wax inside to melt. The wax is then poured out through a tiny hole and liquid metal is poured inside the hollow mould. After the metal has cooled down and solidified, the clay coat is removed and a metal figure of the same shape as the wax figure is obtained. *Even today* the same technique is practiced in many parts of the country.

Examples: Bronze Dancing Girl of Mohenjo-daro, bronze bull of Kalibangan, etc.

The **Dancing Girl** *is the world's oldest bronze sculpture. Found in Mohenjo-daro, this four-inch figure depicts a naked girl* **wearing only ornaments**, *which include bangles on the left arm, and* **amulet** *and* **bracelet** *on the right arm. She stands in a 'tribhanga' dancing posture with the right hand on her hip.*

Terracotta

Terracotta refers to the use of fire baked clay for making sculptures. Compared to the bronze figures, the terracotta sculptures found were less in number and crude in shape and form. They were made using the **pinching method** and have been found mostly in the sites of **Gujarat and Kalibangan (Gujarat)**.

Examples: **Mother Goddess**, mask of horned deity, toys, etc.

The **Mother Goddess** *figure had been found in many IVC sites, which highlights its importance. It is a crude figure of a standing female adorned with necklaces hanging over prominent breasts. She wears a loincloth and a girdle. She also wears a fan-shaped headgear. The facial features are also shown very crudely and lack finesse. She was probably worshipped for prosperity. She might also have been a goddess of fertility sects.*

Bearded Priest *(found in Mohenjo-daro and made of steatite). It is the figure of a bearded man, draped in a shawl with trefoil patterns. The eyes are elongated, and half closed as in meditation.*

Red sandstone figure *of a* **male torso** *(found in Harappa and made of red sandstone). The torso has a frontal posture with well-baked shoulders and a prominent abdomen. There are socket holes in the neck and shoulders, probably for the attachment of head and arms.*

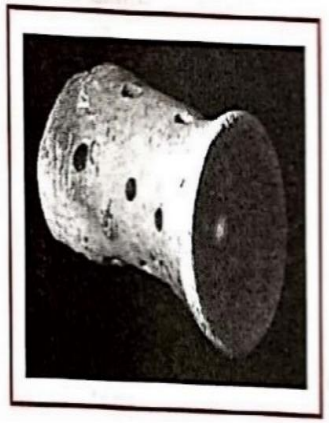
Pottery

The potteries found at the excavation sites can be broadly classified into two kinds – *plain pottery* and *painted pottery*. The painted pottery is also known as **Black and Red Ware** as it used red pigments to paint the background and glossy black paint to draw designs and figures on the red background. Trees, birds, animal figures and geometrical patterns were the recurring themes of the paintings on the Red and Black Pottery.

Red and Black Pottery

Most of the potteries that have been found are very fine **wheel-made wares**, with very few being handmade. Some examples of polychrome pottery have also been found, though rarely. The potteries were used for **three** main **purposes**:

1. Plain pottery was used for household purposes, mainly for storage of grains and water.
2. Miniature vessels, generally less than half an inch in size, were used for decorative purposes.
3. Some of the potteries were **perforated** – with a large hole at the bottom and small holes across the sides. They might have been used for *straining liquor*.



Perforated Pottery

Costumes, ornaments and Jewellery

The ladies were scantily dressed. They wore a short skirt that reached up to the knee; and it was held by a girdle - a string of beads. *Dhoti* wearing was also in practice. Perhaps they did not use footwear. Cotton was used. There is no evidence of linen or wool. Different types of **hairstyles** were used but very long beards were not preferred. The dancing girl from Mohenjodaro has a ponytail. Beards were trimmed and upper lips were shaven. Completely shaven faces with a small beard on the chin are also noticed.

Mirrors of bronze were very common. Men and women both were fond of **ornaments**. Women wore a fan-shaped headdress. Small cones of Gold, Silver, Copper and Faience were worn on the sides of the head. The Forehead was decorated with a **fillet** or a **headband**. **Ear-rings** were made of coils of gold, silver, copper or faience. Probably no nose ornaments were used.

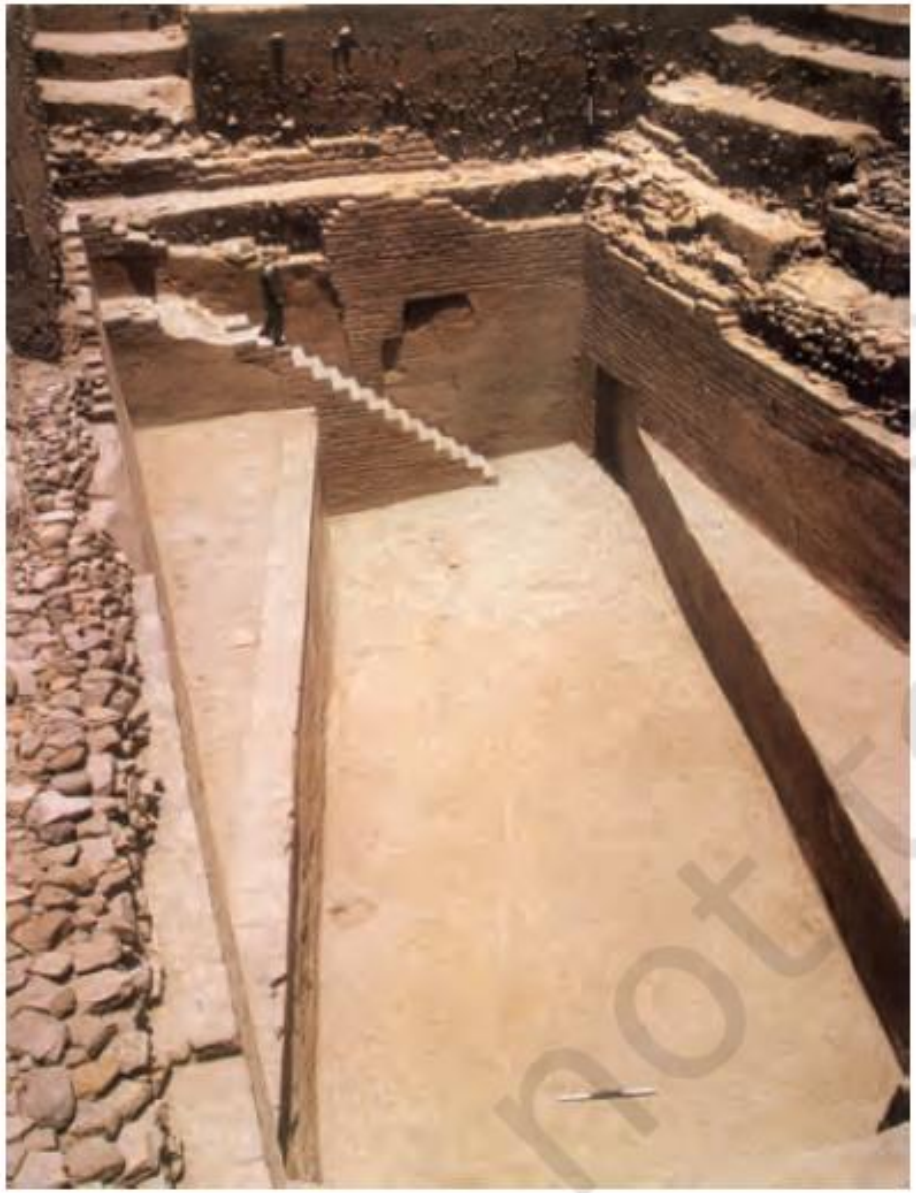
Important Findings (IVC)

Fig. 1.4
Copper tools

➲ Do you think these tools could have been used for harvesting?

Drainage at Mohanjodaro

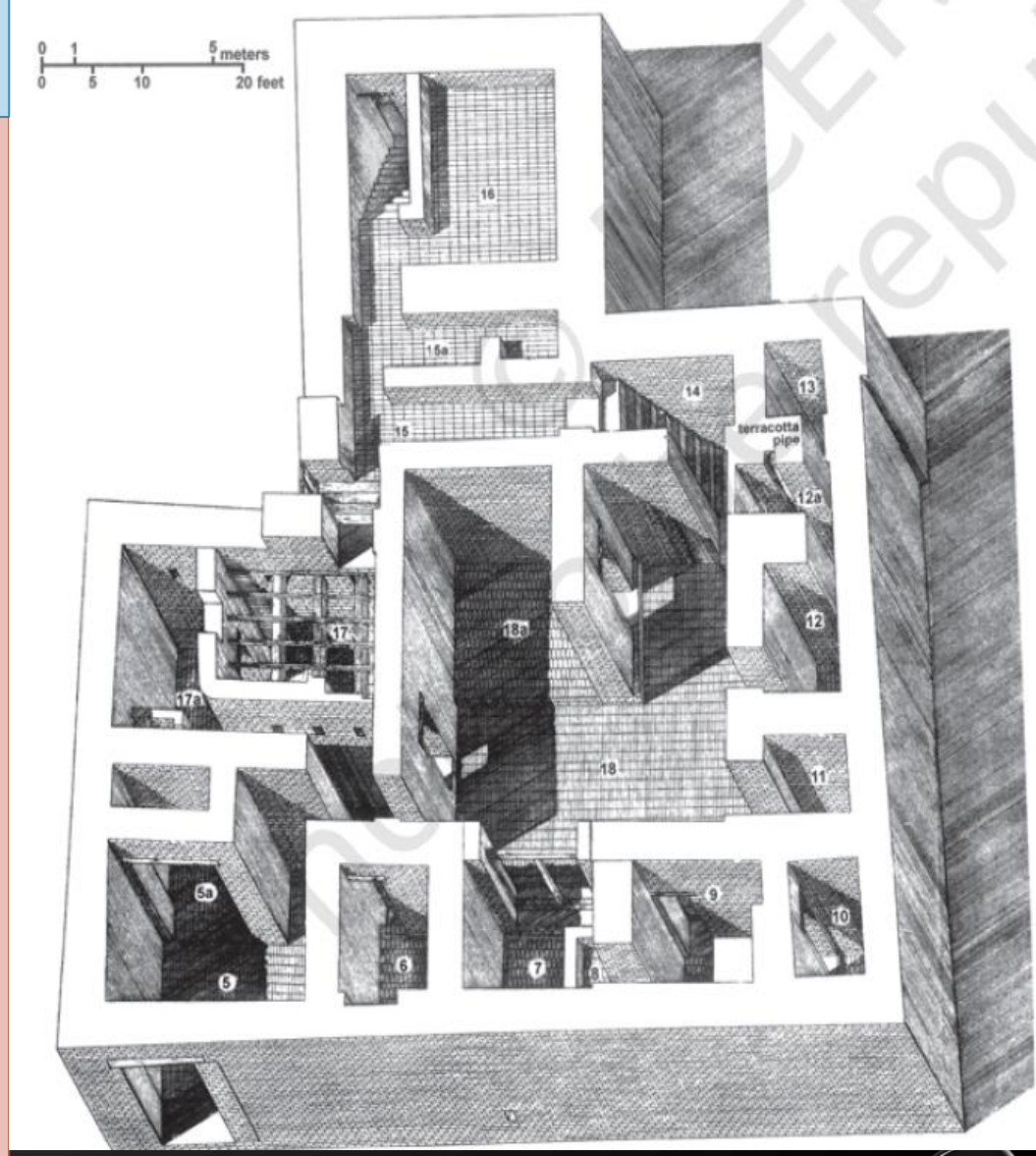
Reservoir at Dholavira
Note the masonry work.



The most ancient system yet discovered

About the drains, Mackay noted: “It is certainly the most complete ancient system as yet discovered.” Every house was connected to the street drains. The main channels were made of bricks set in mortar and were covered with loose bricks that could be removed for cleaning. In some cases, limestone was used for the covers. House drains first emptied into a sump or cesspit into which solid matter settled while waste water flowed out into the street drains. Very long drainage channels were provided at intervals with sumps for cleaning. It is a wonder of archaeology that “little heaps of material, mostly sand, have frequently been found lying alongside drainage channels, which shows … that the debris was not always carted away when the drain was cleared”.

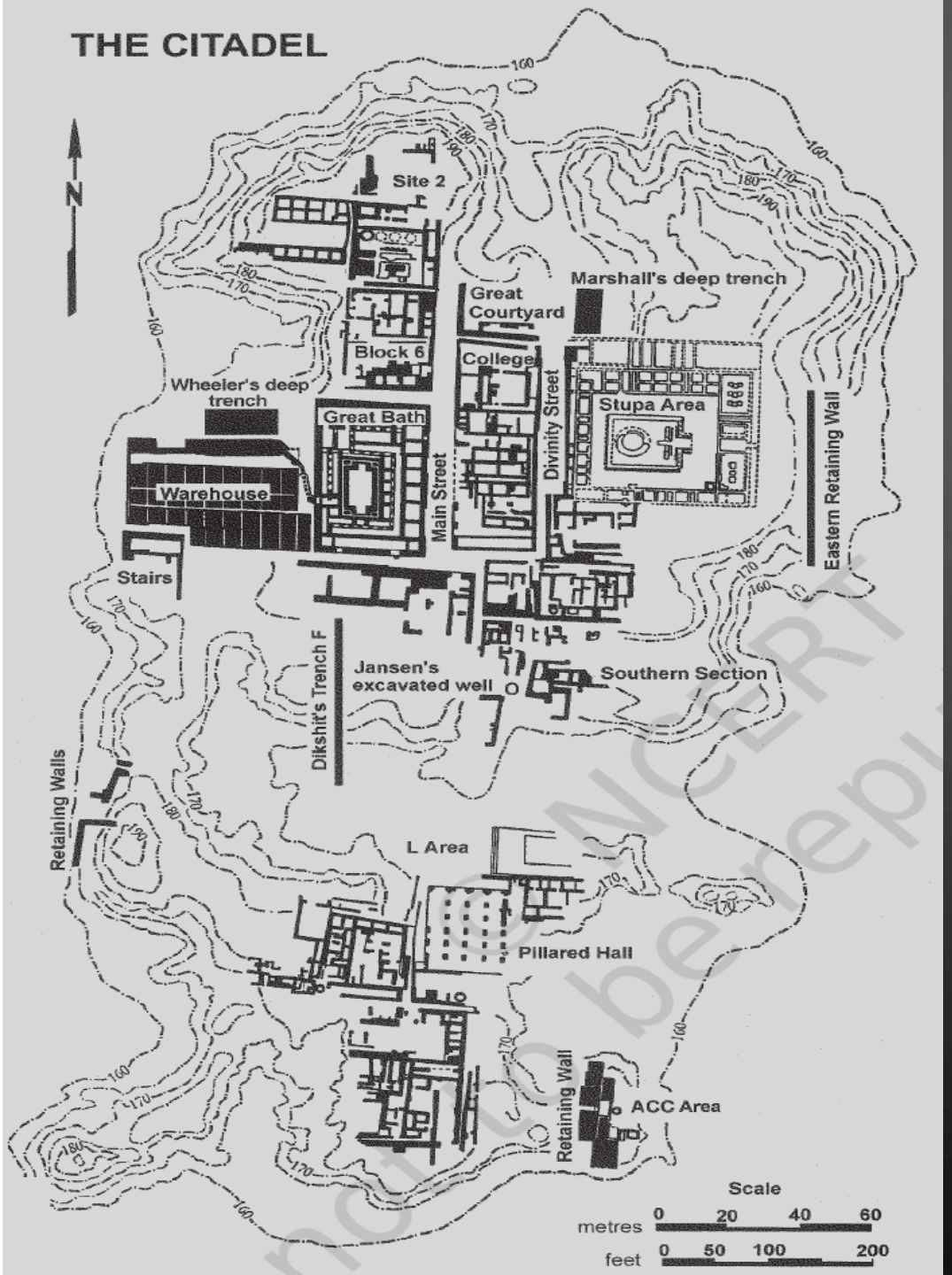
From Ernest Mackay, Early Indus Civilisation, 1948.



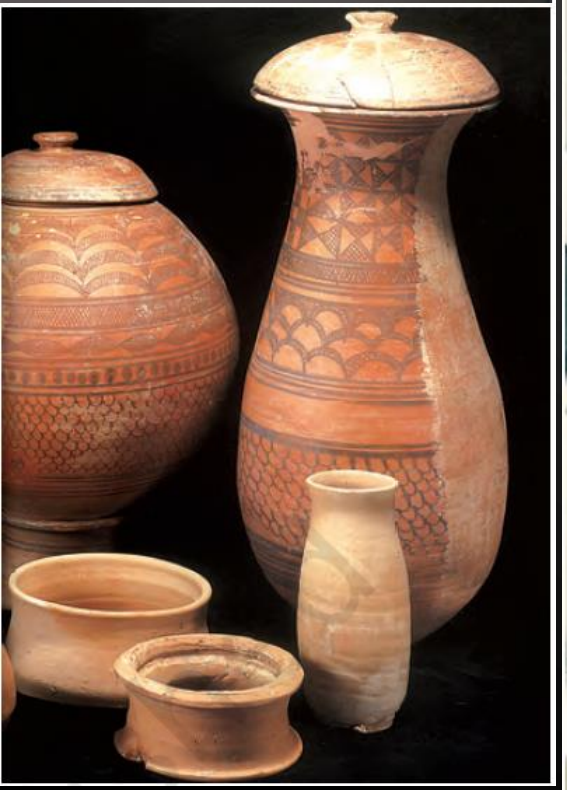
Gamesmen or lingas*?*

This is what Mackay, one of the earliest excavators, had to say about these stones: Various small cones made of lapis lazuli, jasper, chalcedony, and other stones, most beautifully cut and finished, and less than two inches in height, are also thought to be lingas … on the other hand, it is just as possible that they were used in the board-games …

FROM ERNEST MACKAY, Early Indus Civilisation, 1948.

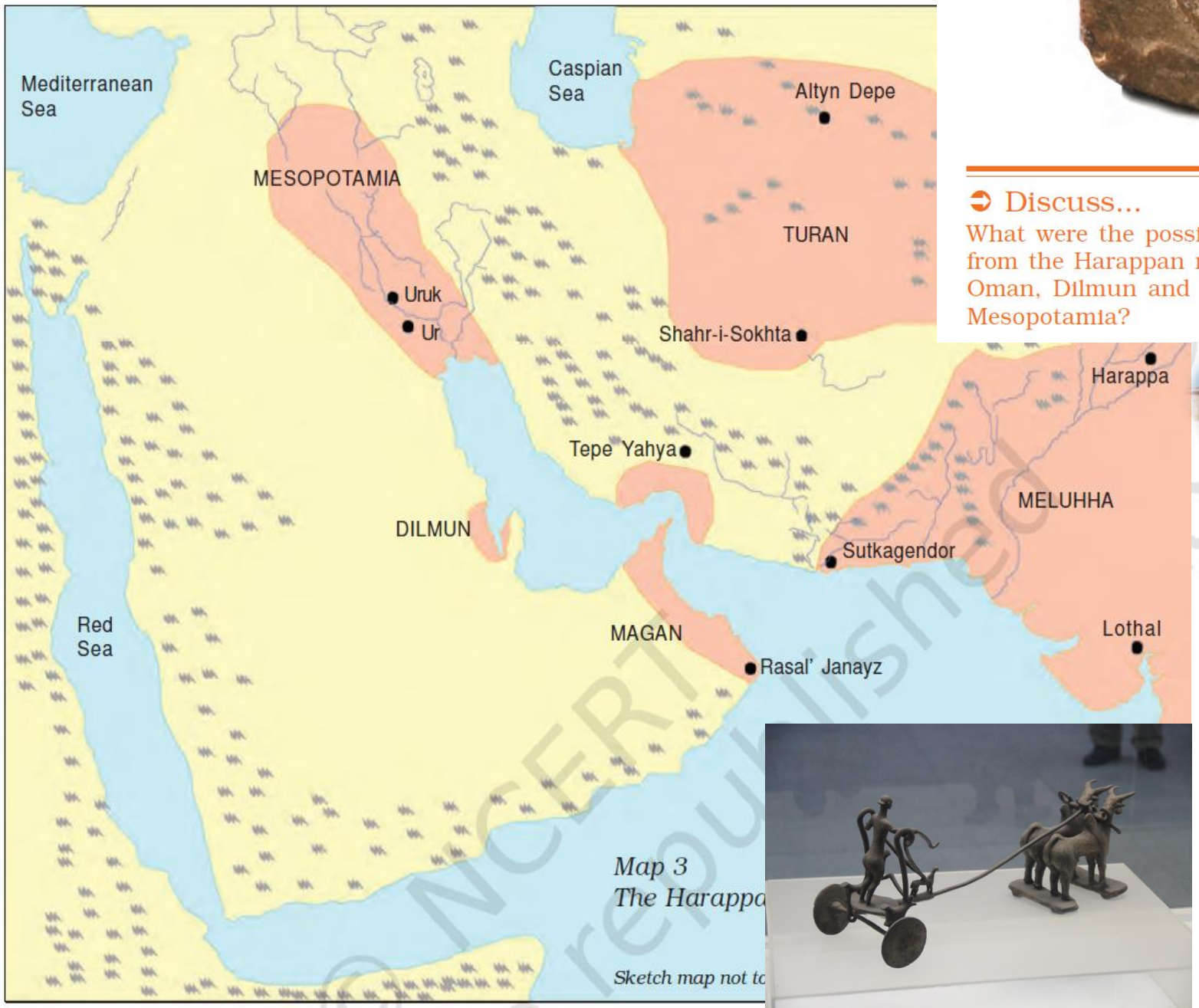

THE CITADEL
N
Site 2
Marshall's deep trench
Great Courtyard
Block 6
College
Wheeler's deep trench
Great Bath
Divinity Street
Stupa Area
Eastern Retaining Wall
Warehouse
Main Street
Stairs
Dikshit's Trench F
Jansen's excavated well
Southern Section
Retaining Walls
L Area
Pillared Hall
Retaining Wall
ACC Area
Scale
metres 0 20 40 60
feet 0 50 100 200

Copper Mirror



Copper & Bronze

Trade Route



➲ Discuss...

What were the possible routes from the Harappan region to Oman, Dilmun and Mesopotamia?

Dockyard at Lothal

Fig. 1.29

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Faience

Unlike stone or shell, that are found naturally, faience is a material that is artificially produced. A gum was used to shape sand or powdered quartz into an object. The objects were then glazed, resulting in a shiny, glassy surface. The colours of the glaze were usually blue or sea green.

Faience was used to make beads, bangles, earrings, and tiny vessels.

Evaluate the significance of seals and sealings in the reconstruction of socio-economic and religious life of the Harappan people.

Aspect	Significance of Seals and Sealings	Examples of Seals
Economic Life	- Trade and Commerce: Seals were used to mark ownership and authenticate goods, indicating a well-developed trade network both within the civilization and with distant regions like Mesopotamia and Persia. They provide evidence of the standardized system of weights and measures.	- Seals depicting a standard script found in Harappa and Mohenjo-Daro. Seals discovered at Lothal indicate maritime trade.
	- Craft Specialization: The variety and intricacy of seals suggest specialized craftsmanship. Different materials (steatite, faience, terracotta) imply a diverse range of artisans.	- Seals made of steatite from Harappa. Faience and terracotta seals from Mohenjo-Daro.
	- Urban Economy: The widespread use of seals in urban centers points to a complex urban economy with administrative control over trade and goods distribution.	- Seals with merchant and trade guild symbols found in Dholavira and Kalibangan.
Social Life	- Social Stratification: The diversity in seal motifs and quality hints at social hierarchies. Some seals may have been exclusive to certain classes or individuals, indicating status.	- High-quality unicorn seals from Mohenjo-Daro, suggesting a higher status.
	- Occupational Identity: Seals likely denoted professional or guild affiliations, helping in understanding the social structure and occupational diversity.	- Seals with depictions of various animals and tools found in Harappa, indicating different professional identities.
Religious Life	- Iconography and Deities: Seals often depict religious symbols, animals, and deities (like the unicorn, bull, and the possible proto-Shiva figure), providing insights into Harappan religious beliefs and practices.	- The "Pashupati" seal from Mohenjo-Daro, depicting a figure seated in a yogic posture surrounded by animals.
	- Ritual Practices: Some seals are thought to have been used in rituals, possibly as amulets or for ceremonial purposes, shedding light on religious rituals and symbolism.	- Seals with fertility symbols and mother goddess figurines from Harappa and Dholavira.

Aspect	Significance of Seals and Sealings	Examples of Seals
Administrative Control	- Record Keeping: Seals and sealings were essential for record-keeping and administrative purposes, indicating a sophisticated system of governance and control.	- Seals with inscriptions and symbols used for administrative purposes found in Harappa and Mohenjo-Daro.
	- Property and Security: The use of seals to secure containers and goods shows an advanced system of property rights and security measures.	- Seals used to mark and secure goods in storage facilities, as found in Harappa.
Communication	- Script and Language: The inscriptions on seals are a primary source for understanding the Harappan script, though it remains undeciphered. They indicate the use of a written language for administrative and possibly ceremonial purposes.	- Seals with inscriptions found in various sites, such as Harappa, Mohenjo-Daro, and Lothal.
Interregional Interaction	- Cultural Exchange: Seals found in Mesopotamia and other regions show the extent of Harappan trade and cultural exchange, highlighting their influence and interaction with other contemporary civilizations.	- Indus seals found in Mesopotamian sites like Ur and Kish.
Daily Life	- Personal Identity: Seals may have been used as personal identification, similar to a signature, revealing aspects of individual identity and personal belongings.	- Personal seals with unique symbols found in Harappa, used to mark personal property or identity.

Water management and its Conservation		
Aspect	**Description**	**Examples/Details**
Drainage System	- The Harappans had an efficient and well-planned drainage system.	- Main drains were covered by corbelled arches made of brick or stone slabs. - Drains for collecting rainwater were separate from sewage chutes and pipes.
Drinking Water and Bathing	- Elaborate arrangements for drinking water and bathing were in place.	- Bathing was done in rivers and in the Great Bath. - Drinking water came from wells, cisterns, etc.
Great Bath	- The Great Bath at Mohenjo-daro is a unique example of water management.	- The floor of the tank was made watertight by finely fitted bricks laid edge to edge with gypsum mortar. - A thick layer of bitumen was laid along the sides of the tank, making it an early example of waterproofing.
Wells	- Wells were a common feature, with a significant number found in various cities.	- Mohenjo-daro had around 700 wells, including both public and domestic wells. - In Allahdino (near Karachi), wells had a small diameter to enable groundwater to rise higher due to hydraulic pressure.
Dockyard at Lothal	- Lothal had an advanced water management system for maintaining water levels.	- The dockyard had a system for maintaining a regular level of water, crucial for its functioning as a trade center.
Water Harvesting at Dholavira	- Dholavira showcased an impressive and unique water harvesting and management system.	- The city had a series of reservoirs and channels to collect and store rainwater, ensuring a sustainable water supply.

To what extent have the urban planning and the culture of the Indus Valley Civilization provided inputs to the present-day urbanisation? Discuss.

IVC Urban Culture Influence

Aspect	Example from India
Urban Layout and Grid System	Chandigarh: Designed with a grid-based layout
Sanitation and Drainage Systems	Dholavira, Gujarat: Elaborate water conservation systems found in Delhi.
Water Management	Dholavira had water reservoirs and tanks. Rani ki Vav, Gujarat can be seen as continuity. Sudarshan Lake
Multi-Functional Buildings	Havelis in Rajasthan: Integrated residential and commercial
Trade and Urban Economy	Mumbai: Economic hub akin to ancient port cities like Lothal
Social Cohesion and Diversity	Kolkata: Rich cultural heritage and diverse population

The ancient civilization in Indian sub-continent differed from those of Egypt, Mesopotamia and Greece in that its culture and traditions have been preserved without a breakdown to the present day. Comment.

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Aspect	Indian Subcontinent	Egypt, Mesopotamia, Greece
Cultural Preservation	Strong continuity due to adaptability and tolerance	Discontinuation due to various factors such as harsh climatic conditions, invasions, and conflicts
Influence of Scriptures	Influence of ancient scriptures like Vedas and Puranas	Influence of religious texts but eventual decline in relevance
Assimilation of Rulers	Assimilation of rulers into Indian culture	Imposition of rulers' culture and subsequent decline
Absence of Institutional Slavery	Lack of institutionalized slavery	Presence of institutionalized slavery contributing to social instability
Sustainability	Sustainable growth due to favorable conditions	Challenges to sustainability due to climatic conditions and external pressures
Factors Contributing to Decline	Absence of harsh climatic conditions, adaptability, and lack of institutionalized exploitation	Harsh climatic conditions, invasions, conflicts, and internal instability

Decline of the Indus Valley Civilization

- Aryan Invasion
- Floods
- Changing of river's course
- Environmental Change
- Over-population
- Decline in Trade

Ramaprasad Chanda
Mortimer Wheeler

Purandhara
Hariyupiya

M. R. Sahni
Robert L. Raikes

Robert L Raikes has taken the case study of Mohenjo-Daro, which shows 7 layers of habitation through stratigraphical analysis which indicates that due to frequent floods; people abandoned the city and again returned when the flood receded.

H. T. Lambrick

Fairservice

Shereen Ratnakar

Continuity of Indus Valley Civilisation

Aspect	Indus Valley Civilization	Continuity and Examples in Later Periods
Religion	- Worship of 'Pashupati Shiva' and 'Lingam'. - Mother Goddess worship.- Sacred tree worship. - Animal worship, including the serpent.- Religious symbols like Swastika.	- Continued worship of Shiva as 'Pashupati' and Lingam in Hinduism. - Durga and other Mother Goddess worship in Hinduism.- Worship of sacred trees like Peepal and Banyan. - Nagas (serpent deities) in Hinduism. - Swastika as a symbol of auspiciousness.
Science and Technology	- Contributions to mathematics, including the numerical and decimal systems.	- Foundation for 'Vedic mathematics'. - The use of the decimal system in later Indian mathematics and worldwide.
Economy and Trade	- Use of punch-marked coins.	- Punch-marked coins continued into the Vedic period and later historical periods, including the Maurya Empire.
Art and Craft	- Pottery making, baked bricks, bead making, jewelry crafting, and manufacturing of various utensils.	- Continued use and advancement in pottery techniques in the Vedic period. - Use of baked bricks in later Indian architecture. - Bead and jewelry making techniques persisted through various periods.
Urban Planning and Architecture	- Advanced urban planning, drainage systems, use of baked bricks for construction.	- Influence on later urban planning in Indian cities. - Continued use of similar drainage systems in ancient Indian settlements. - Baked bricks remained a standard in Indian construction practices

Continuity of Indus Valley Civilisation

Aspect	Indus Valley Civilization	Continuity and Examples in Later Periods
Agriculture and Water Management	- Techniques in irrigation, water storage, and management.	- Techniques adopted and improved upon in later periods, including during the Mauryan and Gupta Empires.
Weights and Measures	- Standardized system of weights and measures for trade.	- Continued use and standardization in later periods, influencing trade and commerce in ancient India.
Writing and Script	- Use of a script that remains undeciphered but indicates early forms of writing.	- Development of later scripts like Brahmi and Kharosthi, potentially influenced by Harappan writing practices.
Social and Cultural Practices	- Practices related to social organization, urban living, and cultural activities.	- Influence on the social structure and urban lifestyle in subsequent Indian civilizations.

Q. 5. Consider the following statements and choose the appropriate one.

i. Technologically, the Chalcolithic stage refers to pre-Harappans but in various parts of the country, it appears after the end of the bronze Harappan culture.

ii. All Chalcolithic cultures are not necessarily pre-Harappan cultures, but all pre-Harappan cultures are necessarily Chalcolithic cultures.

iii. All Chalcolithic cultures are necessarily pre-Harappan cultures, but pre-Harappan cultures are not necessarily Chalcolithic cultures.

(a) i only

(b) i, iii only

(c) i, ii, iii

(d) i, ii only

Q. 13. Consider the following statements regarding the Chalcolithic cultures in India:

i. Some of the Chalcolithic cultures were contemporary with the Harappan culture and others were post-Harappan.

ii. The Chalcolithic culture is characterised by painted pottery, usually red and black.

iii. It was urban as compared to rural Harappan.

Which of the following statement(s) is/are correct?

(a) Only i

(b) i and ii

(c) i and iii

(d) All of the above

Q. 19. Consider the following statements and mark the option which is correct.

i. The first man-made port was found in Harappa.

ii. The main crops of the Indus Valley civilisation were wheat and barley.

iii. The largest Harappan settlement in India is Rakhigarhi in Haryana.

iv. The largest number of settlements are in the Ghaggar-Hakar valley.

(a) i and ii

(b) All of the above

(c) ii, iii and iv

(d) None of the above

Q. 20. Consider the following statements and mark the option which is correct.

i. Rice Husk has been found in Lothal and Rangpur.

ii. Evidence of decorated bricks used in flooring at has been found at Kalibangan.

iii. A clay model of a plough has been discovered from Banawali.

iv. The steatite figure of a bearded man has been recovered from Mohenjo-Daro.

(a) All of the above

(b) ii and iii

(c) None of the above

(d) i, ii and iii

Q. 21. Consider the following statements and mark the option which is correct.

i. The fertility goddess denotes people’s faith in the fertility cult.

ii. The area occupied by the Harappan civilisation was triangular in shape.

iii. Remains of the horse have been discovered from Surkotada.

iv. Painting on a jar resembling the story of the cunning fox from the *Panchtantra* has been found in Lothal.

(a) i and iv

(b) All of the above

(c) None of the above

(d) iii only

Q. 22. Consider the following statements and choose the appropriate options from the answers.

i. Harappan script is regarded as pictographic.

ii. The script was boustrophedon, written from left to right and right to left in alternate lines.

iii. The script of the Harappans has not been deciphered till now.

(a) i, ii, iii

(b) ii, iii only

(c) i, iii only

(d) None of the above

Q. 24. Consider the following statements about the religious practices of the Indus Valley civilisation.

i. Indus people worshipped gods in the form of trees and animals.

ii. People believed in evil forces and used amulets as protection against them.

iii. Pashupati was represented in seals as sitting in a yogic posture.

iv. There is sufficient evidence for the prevalence of phallic worship.

(a) i, ii, iv only

(b) ii, iii, iv only

(c) i, iii only

(d) i, ii, iii, iv only